

## CDS and AMDP Interview Questions:

### 1. Difference between CDS and AMDP ?

| CDS                                                                                                                                                                                                                                 | AMDP                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CDS is a database entity which is used to fetch the data from DDIC/Standard CDS views or any data sources. It also creates the SQL view name in SAP with the name provided as part of annotation<br><b>@AbapCatalog.sqlViewName</b> | AMDP - is a Data base procedure (ABAP Managed Database procedure) is a Global class with the implemented interface (IF_AMDP_MARKER_HDB). It doesn't create any Views in SAP.                                                                                                 |
| Using CDS Variant - CDS Table function, you can do AMDP (which is actually called as AMDP -> ABAP Managed Table function) similar to AMDP.                                                                                          |                                                                                                                                                                                                                                                                              |
| Very Limited functions like CAST, STRING Functions, CURRENCY_CONVERSION, UNIT_CONVERSION, DECIMAL_SHIFT are handled in CDS.<br>Less complex logic is handled at CDS level.<br>Supports CODE PUSH DOWN.                              | AMDP handles the complex logic as AMDP supports only SQL language for Data fetching. Since the AMDP is a Database procedure written at Data base layer (CODE PUSH DOWN), the performance is much higher than traditional ABAP ( data fetching is done at Application server) |
| Debugging CDS view is not possible                                                                                                                                                                                                  | Debugging AMDP is possible                                                                                                                                                                                                                                                   |
| CDS supports very limited SQL.                                                                                                                                                                                                      | AMDP supports 90% of SQL, hence we can use SQL functions like STRING functions, Window functions etc.                                                                                                                                                                        |

### 2. Difference between CDS Table function and AMDP ?

| CDS Table Function                                                                                                                                                                                                                               | AMDP                                                                                                                                                                                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CDS Table function creates a CDS view that is implemented by class method which should implement interface IF_AMDP_MARKER_HDB. The result set of the class method is the CDS view. In this case, it is called as ABAP MANAGED DATABASE FUNCTION. | AMDP - is a Data base procedure (ABAP Managed Database procedure) is a Global class with the implemented interface (IF_AMDP_MARKER_HDB). It doesn't create any Views in SAP. ABAP MANAGED DATABASE PROCEDURE. |
| CDS Table function accepts Parameters that should not exceed 1333 characters. Any ABAP Program that uses CDS table function as data source must convert the Select options to the string and pass as a parameter to CDS Table function.          |                                                                                                                                                                                                               |
| Debugging is possible. As this is similar to AMDP.                                                                                                                                                                                               | Debugging AMDP is possible                                                                                                                                                                                    |

We can create normal CDS Basic Views, Composite views on top of CDS Table function.

AMDP cannot be used with CDS views. AMDP can only be used in ABAP logic to get the data (as a global class method).

### 3. What is Association in CDS View ?

-> Association is used to join the two or more data sources (CDS views or Tables/Views). By default it is Left outer join. Association has Cardinality which denotes the relation (0..1), (1..1), (0..N), (1..N).

-> Association is used to create a Foreign Key Relation ship between data sources with annotation `@ObjectModel.ForeignKey.association` 'Association alias'.

-> Association is an ON DEMAND JOIN. CDS views will never consume the memory from the associated view unless the fields from the associated view are exposed as part of the result set of CDS view.

```
... _scarr.Website,  
... _spfli.AirlineDistance,
```

In this example, the memory is consumed only fro Website and AirlineDistance fields. If the fields are not required then no memory consumption happens.

-> Associations are mainly useful when the extension is carried out on the Standard CDS view, where the requirement is to have text field from Standard CDS view which has associations.

### 4. What are the System Variables in CDS view ?

-> `$session.system_date`, `$session.system_language`, `$session.client`, `$session.user`

### 5. How do you provide security on CDS Views ?

-> To allow only the authorized users to access CDS Views, We need to create DCL CDS Views. Data control Language CDS views will be created on top of Basic CDS views with the authorization object.

```
*[S18] ZCDS_CM_FLIGHT ⌕  
1 @AbapCatalog.sqlViewName: 'ZCSFLIGHT'  
2 @AbapCatalog.compiler.compareFilter: true  
3 @VDM.viewType: #COMPOSITE  
4 @AccessControl.authorizationCheck: #CHECK
```

The highlighted annotation checks for the security. If the authorized user then he/she can access the data.



Access Control  
Views.docx

### 6. How do you expose CDS data source as ODATA Service ?

-> using annotation `@Odata.publish: true` and activate the CDS data object in `/IWFND/MAINT_SERVICE` transaction code.

### 7. How to create a Selection screen for the consumption view ?

-> Consumption view is an Analytic Query with annotation `@Analytic.query : true`

use @Consumption.filter annotation to create selection screen.

8. Can we call any ABAP function module or any abap object in AMDP class ?

-> NO. AMDP is a data base procedure which doesn't support ABAP. It only supports pure SQL.

9. What is the mandatory statement to call class as an AMDP class ?

Implement interface -> IF\_AMDP\_MARKER\_HDB. Hence, class is called as AMDP class.

10. Can we call another AMDP method inside an AMDP method ?

YES.

11. Can we have Normal methods in AMDP class ?

YES.

12. Can we call normal methods in AMDP methods?

NO.